



Arab Academy
for Science, Technology & Maritime Transport

AI-POWERED VIRTUAL HEALTH ASSISTANTS FOR PATIENT ENGAGEMENT

**Case Study: Abu El Rish Japanese Children's Hospital
Cairo University Hospitals – Egypt**

MBA | AI in Business Case Study Analysis Project

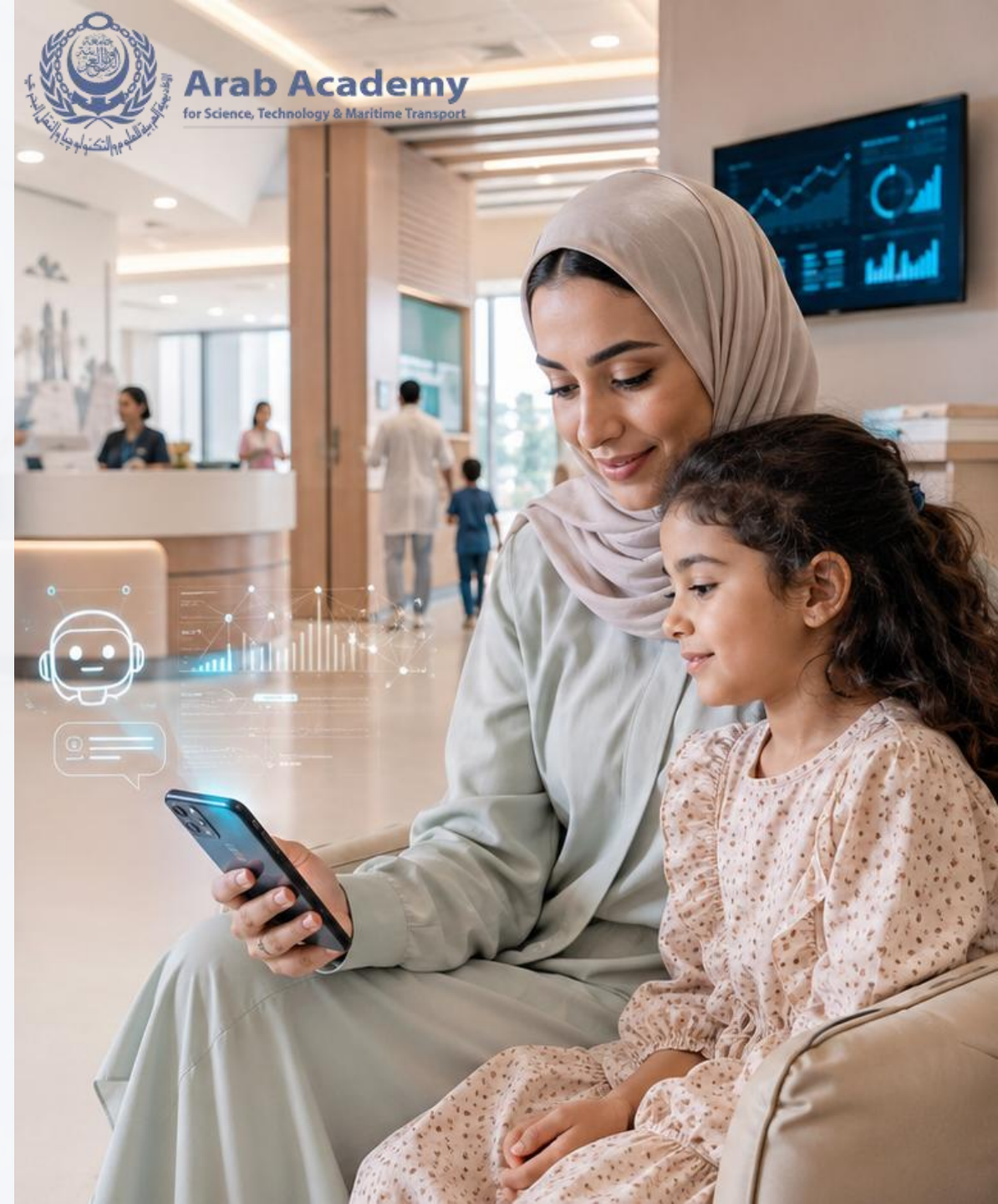
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AI + Analytics

Digital Transformation

ERP + RPA



Case Snapshot — Why Abu El Rish?

A real pediatric-care environment with scale, congestion pressure and an active digital/AI direction

REAL

318 beds

Current 2026 reported capacity of Abu El Rish Japanese Children’s Hospital.

≈500K children / year

Cairo University reports roughly half a million children annually across its pediatric hospitals.

72% occupancy

Reported occupancy at Abu El Rish Japanese in July 2025.

Waiting-list pressure

Outpatient expansion was explicitly linked to reducing waiting lists and congestion.

BUSINESS CONTEXT



Case thesis

Use a governed Virtual Health Assistant to make caregiver engagement more accessible, personalized and data-driven — while keeping clinical decisions under human control.

Sources: Cairo University / Faculty of Medicine (2026 hospital development; 2025 occupancy; outpatient development and waiting-list statements).

Problem Statement & Research Question

The case focuses on patient/caregiver engagement — not autonomous diagnosis

Business problem

- Large pediatric volumes create repeated information, navigation and follow-up needs.
- Caregivers interact across appointments, clinics, referrals, results, documents and hospital guidance.
- Manual / fragmented interactions can consume staff time and create avoidable friction.
- The hospital needs scalable digital self-service without compromising child safety or privacy.

Research question

How can an AI-powered Virtual Health Assistant improve patient and caregiver engagement at Abu El Rish through conversational AI, analytics, ML, RPA and enterprise-system integration while maintaining safety, privacy and human oversight?

Scope boundary

IN SCOPE

Navigation • appointments • reminders • pre-visit guidance • status tracking • human escalation

AI / ANALYTICS

NLP • supervised ML • segmentation • semi-supervised intent classification • prescriptive actions

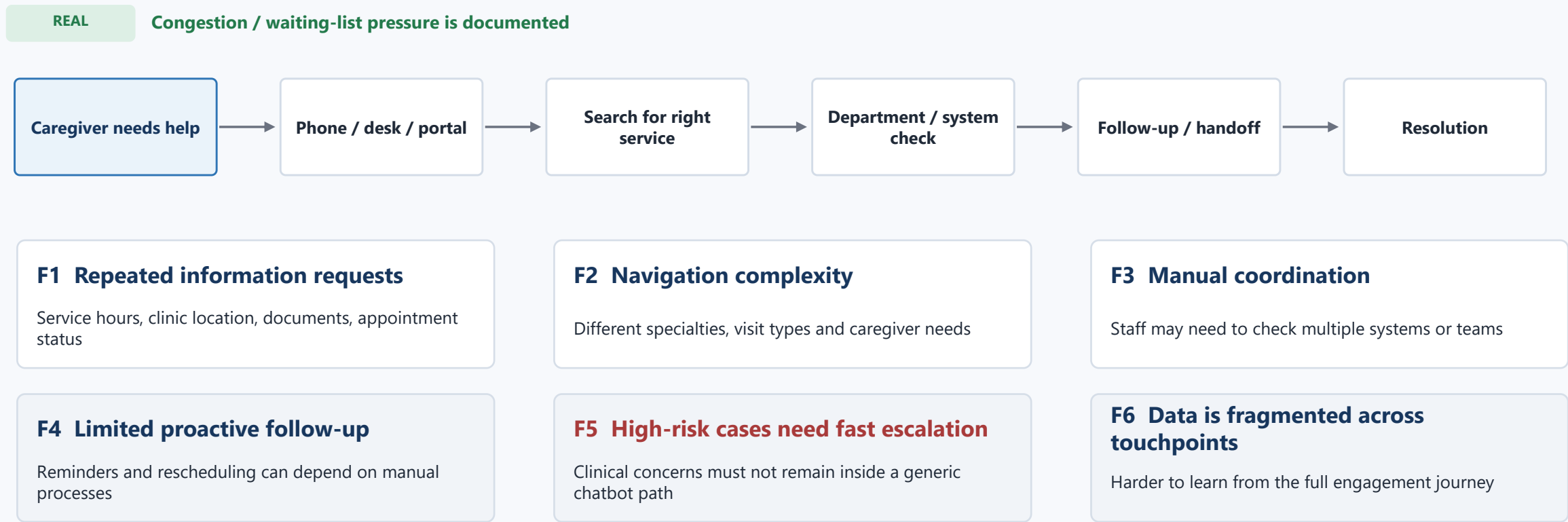
OUT OF SCOPE

Autonomous diagnosis • prescribing • unsupervised clinical advice

Human-in-the-loop is mandatory

Current-State Journey — Where Engagement Friction Appears

Evidence-backed congestion pressure + analytically modeled engagement pain points



Source for real problem context: Cairo University outpatient development statements explicitly cite reducing waiting lists and outpatient congestion. Other friction points are case-study analytical hypotheses.

Data Foundation — Real Evidence + Transparent Simulation

The case separates published facts, derived values and synthetic analytical data

REAL

- 318 beds
- ≈500K children/year
- 72% occupancy (Jul 2025)
- Documented outpatient congestion pressure

DERIVED

- Average daily volume from annual figure
- Business ratios calculated from published values
- No new “official” facts created

SIMULATED

- 10,000 engagement records
- Waiting time / no-show / satisfaction
- VHA use / reminders / escalation
- Used for MBA analytics & ML

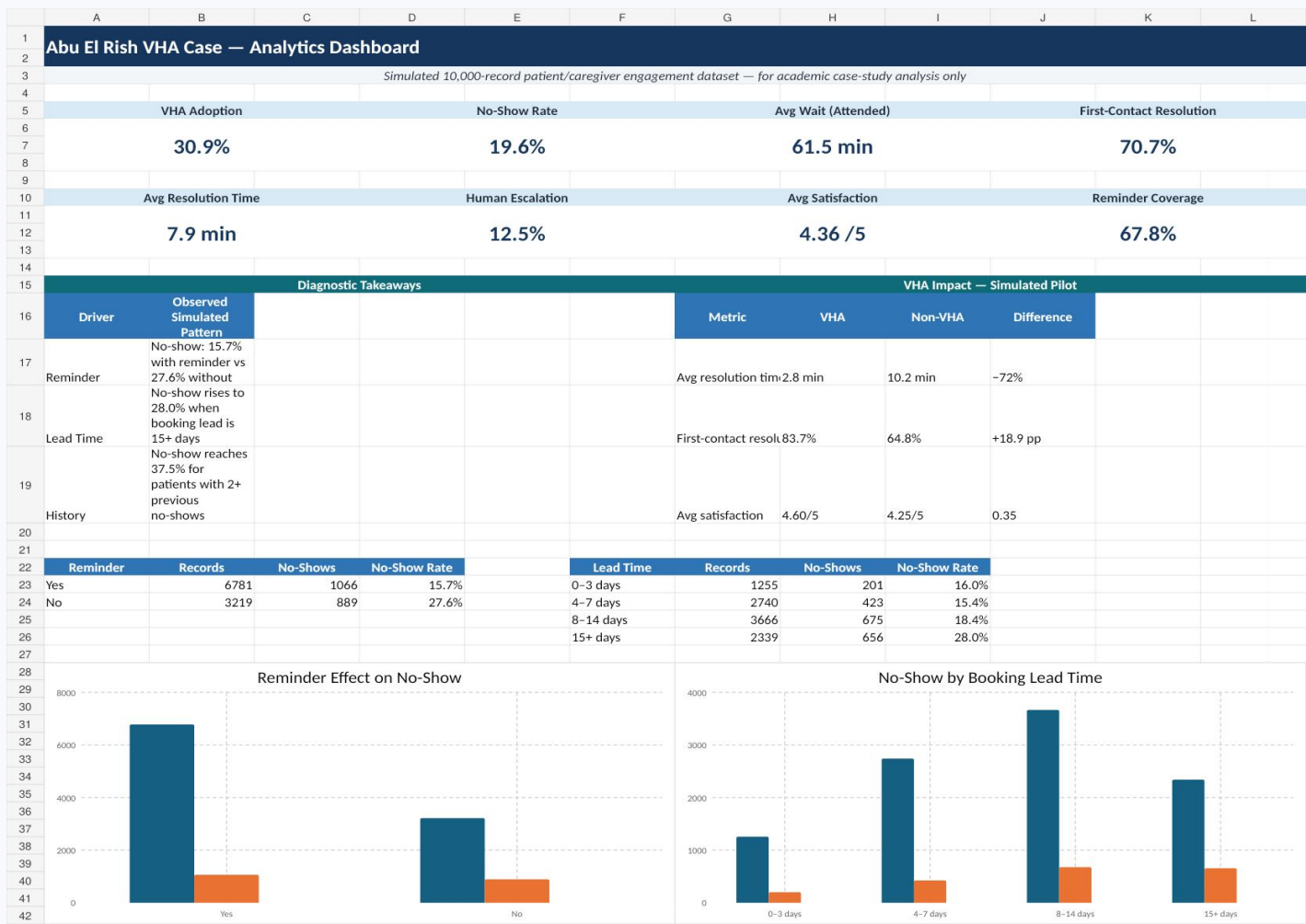
Simulated interaction schema



Simulation is explicitly allowed by the MBA syllabus when clearly stated. All model outputs are labeled simulated and are not presented as actual Abu El Rish performance.

Descriptive Analytics — What Is Happening?

10,000 simulated patient/caregiver engagement interactions



Baseline simulated KPIs

- 30.9%

VHA adoption
- 19.6%

Overall no-show
- 61.5 min

Average attended wait
- 70.7%

First-contact resolution
- 4.36 / 5

Average satisfaction

Purpose: establish a measurable baseline

Source: simulated academic dataset developed for this case study; not actual Abu El Rish operational data.

Diagnostic Analytics — Why Are No-Shows Higher?

The simulation embeds plausible drivers so the analysis can test cause-oriented relationships

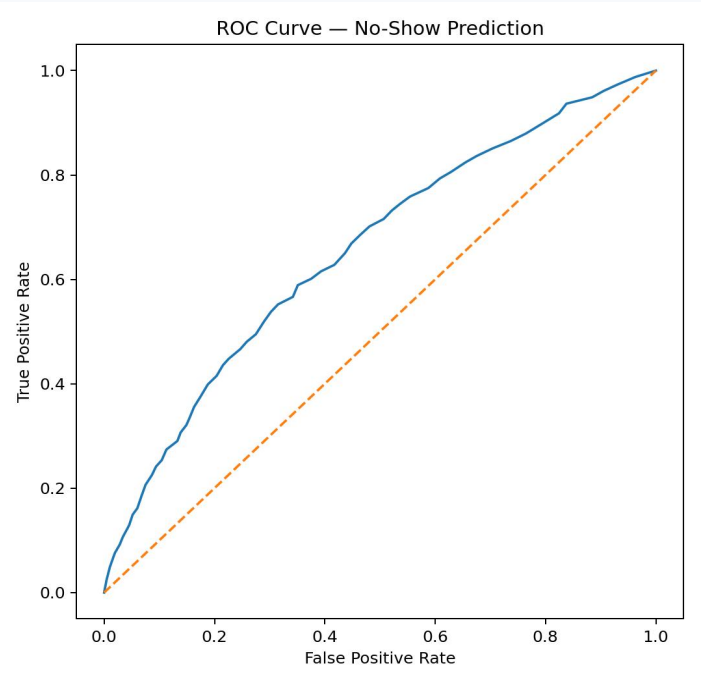
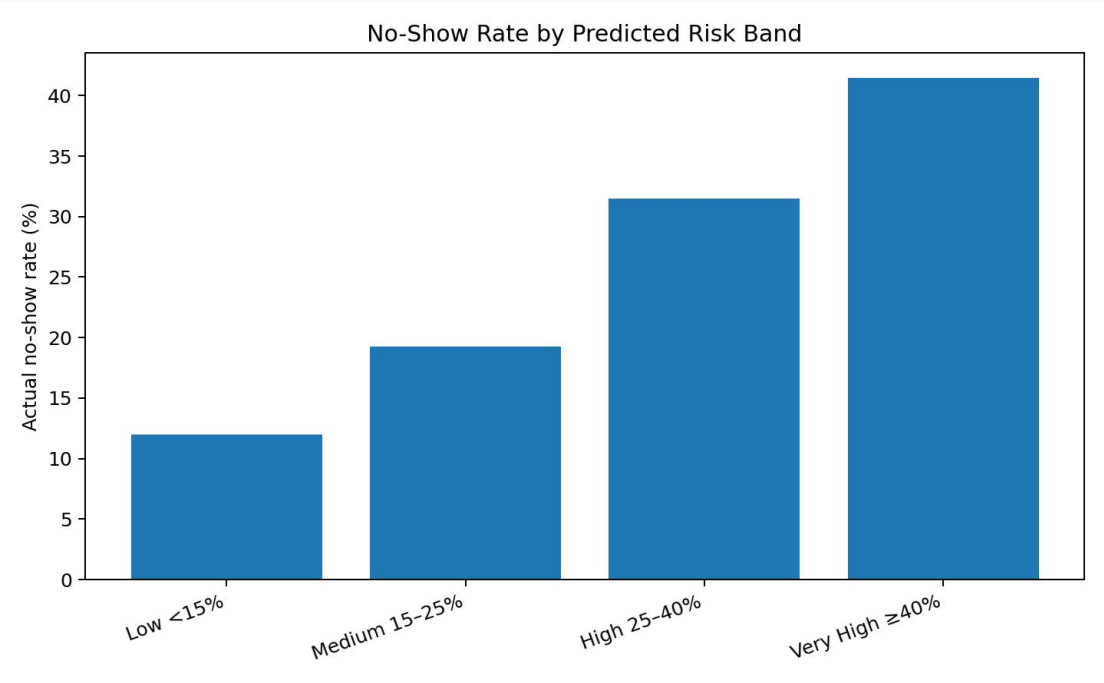


Diagnostic conclusion: reminders, lead time and prior attendance history are useful variables for predictive targeting; VHA is best positioned as a low-cost intervention channel.

Source: simulated academic dataset developed for this case study.

Predictive Analytics — Supervised ML for No-Show Risk

Transparent proof-of-concept model: Logistic Regression on a 75/25 train-test split



Model summary

0.655

ROC AUC

39.6%

Top 10% actual no-show

≈2.0×

Top 10% lift

47.6%

Recall when flagging top 30%

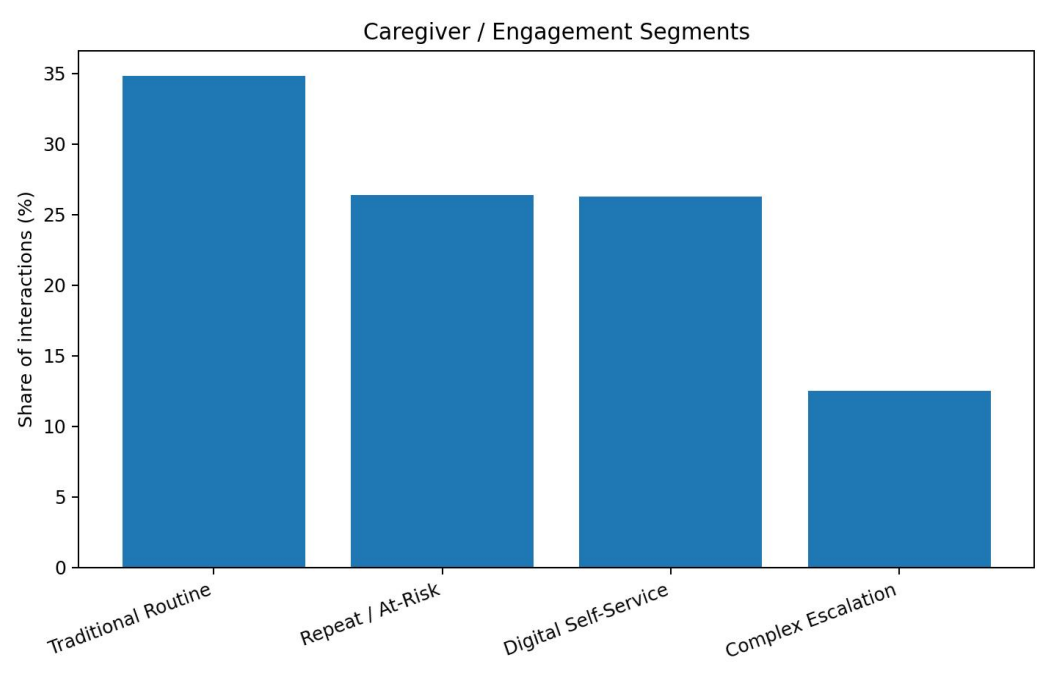
Managerial interpretation

The model is not “perfect,” but it ranks risk well enough to concentrate intervention resources on patients most likely to miss appointments — a suitable decision-support use case.

Source: simulated academic dataset. Model outputs are proof-of-concept only and require validation on real hospital data before operational use.

Unsupervised + Semi-Supervised Learning

Segment journeys today; learn from partially labeled caregiver messages over time



Traditional Routine 34.8%
Promote digital self-service

Repeat / At-Risk 26.4%
Target reminders + proactive confirmation

Digital Self-Service 26.3%
Keep digital-first; automate follow-up

Complex Escalation 12.5%
Fast context-rich human handoff

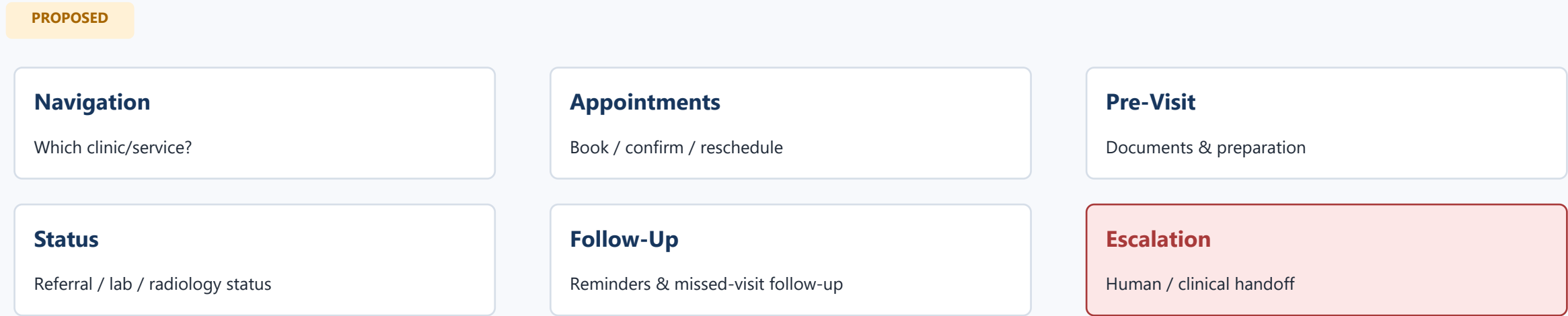
Semi-supervised intent classification



Segmentation and semi-supervised design are based on the simulated case dataset and conceptual NLP design.

AI Solution Overview — The Virtual Health Assistant

A patient/caregiver engagement layer combining NLP, analytics, decision support and human escalation



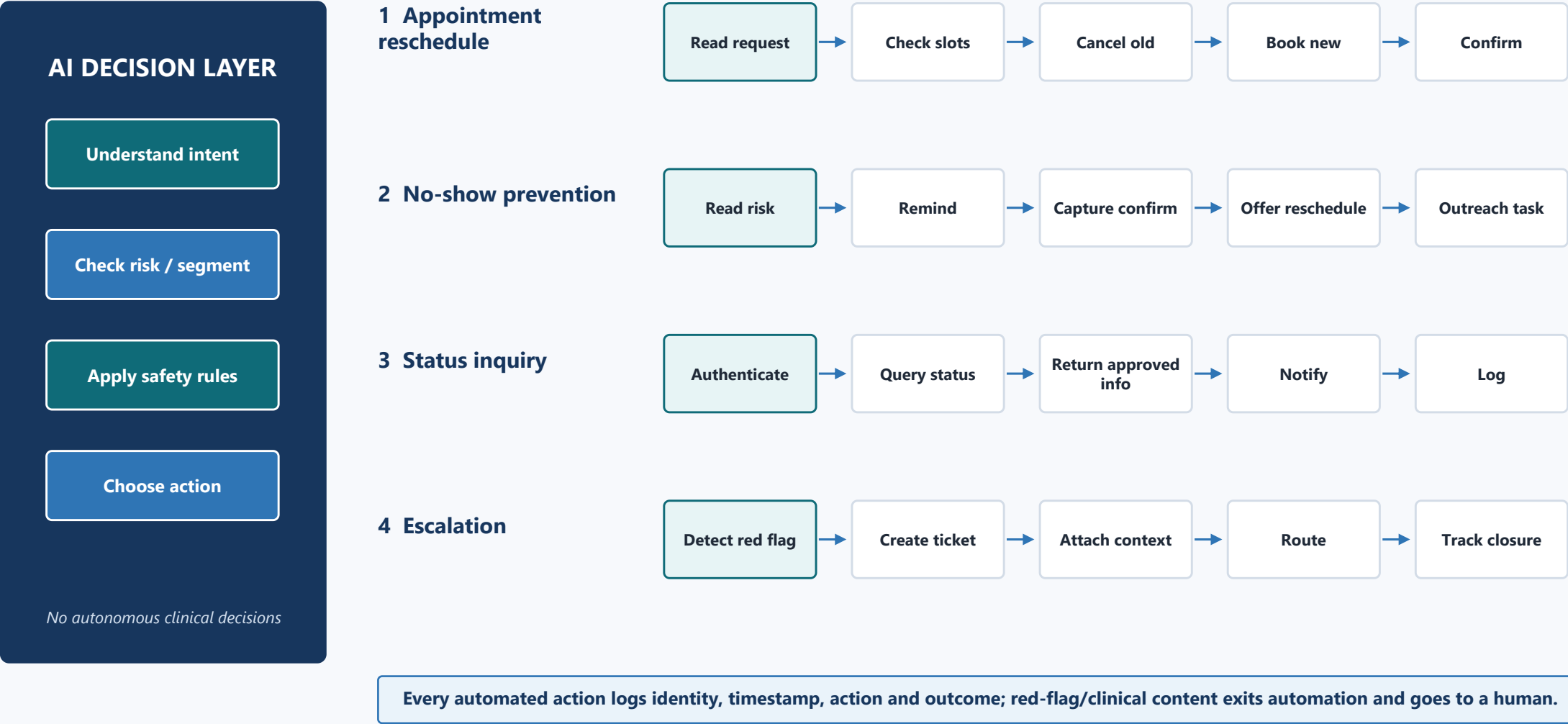
CONVERSATIONAL AI + NLP → ML / ANALYTICS → RPA / APIs → HOSPITAL / ERP SYSTEMS

Safety boundary The VHA supports engagement and administrative navigation; it does not independently diagnose or prescribe. Clinical/red-flag content is escalated.

SDLC Delivered via an iterative/Agile Software Development Life Cycle (SDLC): Plan → Analyze → Design → Develop → Test → Deploy → Maintain, with versioned, approved model/content releases.

RPA — Turning AI Decisions into Enterprise Actions

AI interprets; RPA / APIs perform repeatable transactions with auditability



Digital Transformation — Before vs After

The value comes from redesigning workflows, not merely adding a chatbot

BEFORE	AFTER
Reactive Caregiver initiates every interaction	Proactive Risk-based reminders and follow-up
Fragmented Information spread across touchpoints	Orchestrated VHA connects systems through APIs/RPA
Manual Staff checks and follow-ups	Automated routine work Humans focus on exceptions
One-size-fits-all Same process regardless of risk	Personalized Action varies by segment/risk
Limited learning Little closed-loop analytics	Data-driven Dashboards + model monitoring + learning

Transformation outcome: accessible engagement + lower administrative burden + better appointment utilization + stronger management insight

Context: Cairo University / Kasr Al Ainy publicly describes AI as moving toward institutional application in medical education, clinical and research environments (2026).

ERP + HIS Integration — Enterprise Data Flow

Vendor-neutral conceptual architecture; interoperability should use controlled APIs and healthcare data standards where applicable

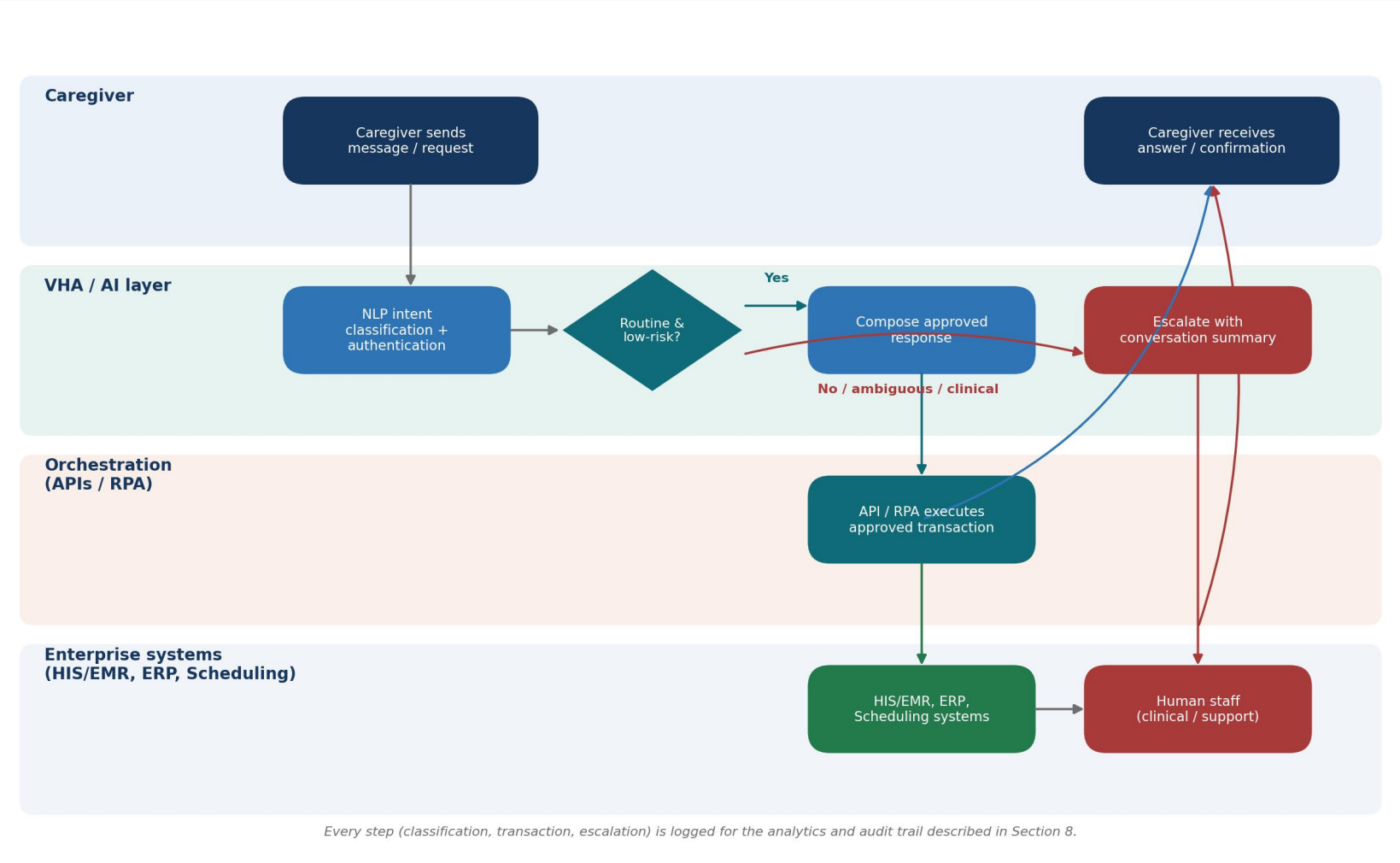
Enterprise module	Data / transaction	VHA / AI use	Business value
Finance	Billing / charges / cost center	Approved administrative queries	Fewer manual lookups
HR / Workforce	Staff availability / roles	Task routing + role-based access	Faster handoffs
Procurement / Inventory	Stock / supply status	Non-clinical availability workflows	Better resource coordination
Operations	Clinic capacity / slots	Appointment & queue decisions	Higher utilization
HIS / EMR	Patient / visit context	Contextualized guidance	One patient journey

VHA request → identity / consent → API / RPA → enterprise system → approved response / transaction → analytics + audit trail

Interoperability reference: HL7 FHIR is a standard for electronic healthcare data exchange; Microsoft healthcare documentation also uses FHIR-based data models. Specific Abu El Rish vendors are not asserted.

ERP Integration — Process Flow (BPM View)

Swimlane view of the caregiver / VHA / orchestration / enterprise-systems workflow described on the previous slide



Business Impact & Illustrative ROI

Financial assumptions are simulated; operational effects are linked to the case analytics

YEAR-1 ROI

≈ 26%

$ROI = (Benefit - Cost) \div Cost \times 100$
 $= (1.955M - 1.550M) \div 1.550M \times 100$

Benefit = gross annual benefit; Cost = Year-1 investment (EGP)

PAYBACK PERIOD

≈ 9.5 mo

$Payback = Investment \div (Benefit \div 12)$
 $= 1.550M \div (1.955M \div 12)$
Months to recover the Year-1 investment from monthly benefit

3-YEAR ROI

≈ 139%

$ROI_3 = (\sum Benefit_{3yr} - \sum Cost_{3yr}) \div \sum Cost_{3yr} \times 100$
 $= (5.865M - 2.450M) \div 2.450M \times 100$
Basis: EGP 1.955M/yr benefit (flat); cost = 1.550M (Y1) + 0.45M + 0.45M (Y2–Y3 maintenance)

WHERE THE VALUE COMES FROM

Patient / Caregiver

Faster, 24/7 self-service access

Operational

Lower resolution time, higher FCR

Financial

Avoided admin cost + recovered visits

Strategic

Reusable AI + data platform for scale

ANALYTICS-LINKED PROOF POINTS

10.2 → 2.8 min

Resolution time

64.8% → 83.7%

First-contact resolution

27.6% → 15.7%

No-show (no reminder → reminder)

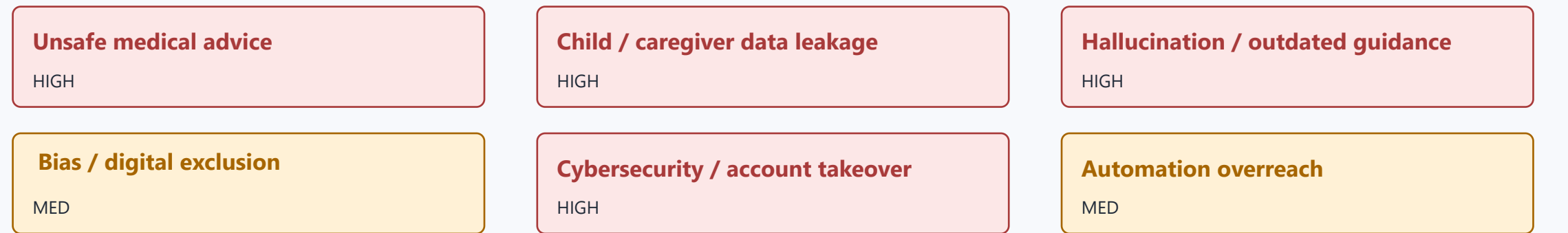
4.25 → 4.60 / 5

Caregiver satisfaction

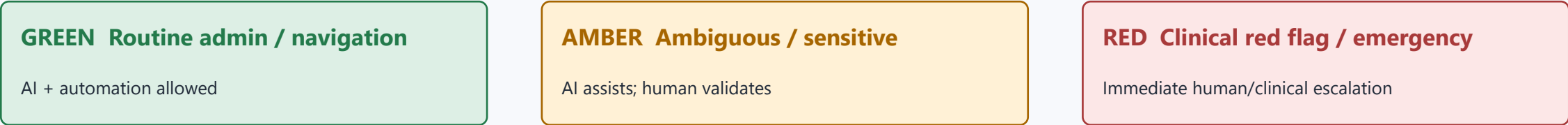
Scenario assumptions for academic evaluation — not Abu El Rish financial results or approved budgets.

Risks, Ethics, Privacy & ISO/IEC 27001

Healthcare AI must be governed as a sociotechnical system — not simply as software



Human-in-the-loop operating rule



ISO/IEC 27001:2022 — Clause 6.1.2 risk assessment



Egypt compliance overlay: Personal Data Protection Law No. 151/2020 + Executive Regulations No. 816/2025; privacy, consent, access control and auditability must be built into the solution.

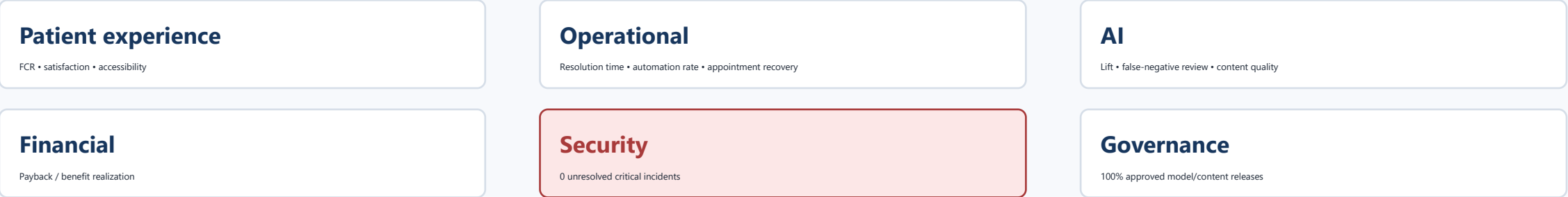
References: WHO Ethics & Governance of AI for Health; ISO/IEC 27001:2022; Egypt PDPC legal framework.

Implementation Roadmap & Scale Gates

Start narrow, prove value and safety, then expand



Scale / no-scale gates



Safety override: any critical patient-safety or information-security failure → STOP, remediate, revalidate before further scale.

Final Recommendation

Proceed as a governed digital-transformation program — not a stand-alone chatbot project

RECOMMENDATION: LAUNCH A CONTROLLED PILOT



Expected strategic outcome

More accessible pediatric journeys • lower administrative burden • better appointment utilization • scalable data-driven engagement

Selected references: Cairo University / Faculty of Medicine (2025–2026); WHO (2021); Egypt PDPC; ISO/IEC 27001:2022; HL7 FHIR; JMIR / Journal of Medical Systems systematic reviews on conversational agents; Entropy systematic review on patient no-show prediction.

Thank you